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Policy Analysis

Incorporating Health Cobenefits into Province-Driven Climate Policy: A Case of Banning New Internal Combustion Engine Vehicle Sales in China

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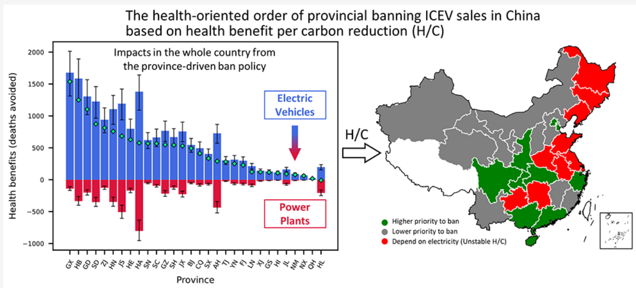
ABSTRACT: Incorporating health cobenefits from coabated air pollution into carbon mitigation policy making is particularly important for developing countries to boost policy efficiency. For sectors that highly depend on electrification for decarbonization, it remains unclear how the increased electricity demand and consequent health impacts from sectoral mitigation policy in one province would change the scale and the regional and sectoral distributions of the overall health impacts in the whole country. This study chooses the banning of new sales of internal combustion engine vehicles in the private vehicle sector in China as a case. The results show that, without carbon neutrality and air pollution control goals in electricity generation, 53% of CO_2 reduction and 65% of health benefits from the private vehicle sector would be offset by increased electricity demand. The regional distributions of CO_2 reduction and health benefits due to a province-driven ban policy are greatly uneven, as the top five provinces take up over one-third of the total impact in China. Health benefits per ton of carbon reduction (H/C) may vary by up to 8 times across provinces. Finally, the provinces in southeast China and the Sichuan Basin, with their stably high H/C values, are suggested to enact the province-driven ban policy first.



KEYWORDS: *carbon mitigation, health cobenefits, private vehicle, province-driven policy, policy implementation order*

1. INTRODUCTION

To reach the 2 and 1.5 °C temperature rise limit goals strengthened in the Glasgow Climate Pact,[1] global CO_2 emissions must quickly approach net zero by approximately 2050.[2] For developing countries that are facing multiple challenges from carbon reduction, air pollution control, and public health improvement,[3−6] it is particularly important for them to incorporate health cobenefits from coabated air pollution into climate policy-making so that the efficiency of climate policy can be greatly boosted.[7−9] One approach to make a health-oriented climate policy is to rank the regions or emission sources in the order of highest health cobenefits per ton of carbon reduction to the lowest, with the aim of suggesting a policy implementation order at the regional or emission source level.[10] This approach has been used in several previous studies, particularly in the electricity generation sector.[10−13] For the transport sector, which accounts for 23% of global energy-related CO_2 emissions and is expected to increase more rapidly in developing countries,[14] existing studies only quantified the health cobenefits of decarbonizing the transport sector under different scenarios[15−19] and did not explore how to optimize climate policy with the aim of maximizing the health cobenefits.[20] In this study, we choose private vehicles in the road transport sector in China as an example to incorporate health cobenefits into province-driven carbon mitigation actions and further suggest the order of provincial policy implementation.

Banning new sales of internal combustion engine vehicles (ICEVs) (hereinafter referred to as "the ban policy"), recently highlighted in COP26,[21] is a promising deep-decarbonization measure in the private vehicle sector in the postsubsidy era. To date, Hainan (located in the southern part of China) has been the pioneering province to announce a ban on the sale of ICEVs by 2030, and the schedule of the ban policy in other provinces is under hot discussion.[22−24] Therefore, discussing how to arrange an appropriate banning order at the provincial level has practical significance in the policy context of China. As electrification is a widely recognized pathway to replace ICEVs in the private vehicle sector in China,[25−28] the health

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cobenefits of provincial-level ban policies are mainly the trade-offs between two aspects: the health benefits contributed by the phasing-out of ICEVs and the health damage brought by the new sales of electric vehicles (EVs) to fulfill transportation needs.[15,16,19,29] The former aspect (which results in the reduced use of gasoline or diesel) is well-understood, while the latter needs further explanation. The ban policy would lead to an increased number of EVs and a higher demand for electricity generation,[30−32] so there would be sectoral and regional transfers in the effect of the ban policy. For sectoral transfers, before electricity becomes completely carbon neutral and clean, the electrification process essentially transfers CO_2 and the amount and species of air pollutant emissions from the transport sector to the electricity generation sector. With the progress of pursuing carbon neutrality targets and the determination to improve air quality in China, the energy mix and strength of air pollutant emission control in each province will change dynamically, which will both affect the interregional electricity transmission regime and cause emissions and health impacts. By assuming the electricity generation and emission as several fixed constants, existing studies have not taken the dynamic changes of the electricity sector into consideration resulting in a bias in future projections. Consequently, the separate contributions of carbon mitigation and air pollution control actions in the electricity sector when implementing electrification in the end-use sector remain unclear. For regional transfers, due to the rapid development of interregional electricity transmission, the electricity charged by vehicles in a certain province may be generated locally and, at the same time, imported from other provinces. Therefore, the CO_2 and air pollutants and the related health benefits brought about by the ban policy in one province (a province-driven policy) could also spill over to other sectors and provinces through electricity consumption and interregional electricity transmission. Although we already know the health impacts of increased numbers of EVs in the whole of China, even in high spatial resolution,[15,16,19] the size of each province-driven policy effect that is brought to the whole of China remains unclear when considering the electricity generated in both the province itself and the electricity-exporting province.[33] Hence, there is an urgent need to quantify the carbon emissions and health benefits in the transport and electricity sectors due to the province-driven ban on the sale of ICEVs instead of the aggregated impacts from all provinces in a country, taking into consideration future dynamic changes in the energy mix, electricity transmission, carbon mitigation, and air pollution control, with the aim of supporting climate policy design to maximize health cobenefits.

To fill the aforementioned gaps, an integrated assessment framework with energy, environment, and health models was built to quantify the carbon emission and health cobenefit changes due to a province-driven ban policy in China. We took private vehicles as an example because they are the major CO_2 emitters in road passenger transport and are of great concern to the public and the market.[21,34] Specifically, three subquestions are designed to be addressed: (1) How much CO_2 reduction and health benefits would be generated by the ban policy under different carbon mitigation and air pollution control actions in the electricity sector? (2) What are the regional and sectoral distributions of CO_2 reduction and health impact changes due to a province-driven ban policy? (3) How much health benefits per carbon reduction (H/C) due to a province-driven ban policy are in each province? Finally, we determined a ranking based on the H/C for our suggestion of the provincial order of ban on the sale of ICEVs.

2. METHODOLOGY

2.1. Modeling Framework and Scenario Design. For the accounting system boundary, the electricity generation increased by electric vehicles in the electricity sector and pump-to-wheel (PTW) periods of vehicles in the transport sector were considered in this study. The other upstream processes (such as manufacturing, coal mining, oil refining, etc.) were not included since, currently, no relevant research can answer where they take place explicitly driven by a specific province, and thus, they would not affect the provincial order by using the national uniform parameters. We considered 30 provinces (Tibet excluded due to lack of data) of mainland China, and the time span was from 2015 to 2050. In this study, we developed a bottom-up modeling framework following the structure of "Energy-Emission-Health" used by most studies in the area (Figure S12). The Chinese Provincial Passenger transport Energy and Emission (CPPEE) model[23] and Multiregional model for Energy Supply system and their Environmental ImpaCts (MESEIC) model[35−38] were employed to estimate the future private vehicle (PV) fleet size, electricity generation and transmission, and energy demand and emission. Next, the quasi-input−output (QIO) model[39] was applied to calculate the indirect impacts through interprovincial grid transmission of electricity in each province. Then, the $PM_{2.5}$-related ($PM_{2.5}$ = tiny particulate matter or droplets in the air that are 2.5 μm or less in width) health impact assessment was conducted by linking the localized reduced complexity chemical transportation model (RCM) and the global exposure mortality model (GEMM).[40] Finally, the suggestion of provincial banning order according to the health benefits per carbon reduction (H/C) ranking was put forward.

To better understand the factors for carbon reduction and the health impacts of a province-driven ban on ICEV sales in each sector, we developed multiple scenarios for the PVs in the transport sector and the electricity sector (Table 1). For PVs, the reference (REF) and ban policy (BPS) scenarios were set for future PV fleet change estimations in the CPPEE model. The REF scenario is intended to serve as a counterfactual reference for understanding any additional policy impacts and assumes that future penetration of all vehicles remains constant at the base-year levels. The BPS is the policy scenario in which new sales of ICEVs and plug-in hybrid electric vehicles (PHEVs) are banned and replaced by battery electric vehicles (BEVs) in 2035 since the BEV is the most possible and

Table 1. Scenario Settings of This Research[a]

PV Scenario	Electricity Scenarios Matrix				
BPS minus REF (ban policy)			Air Pollution Control		
			Base	CLE	MFR
	Carbon Mitigation Target	BAU	BAU-Base	BAU-CLE	BAU-MFR
		NET	NET-Base	NET-CLE	NET-MFR

[a]BPS − REF is the difference between BPS and REF, to represent the effect of the ban policy. In the electricity scenarios matrix, the BAU and NET represent the scenarios without and with carbon mitigation targets (near zero); and BASE, CLE, and MFR represent different levels of air pollution control.

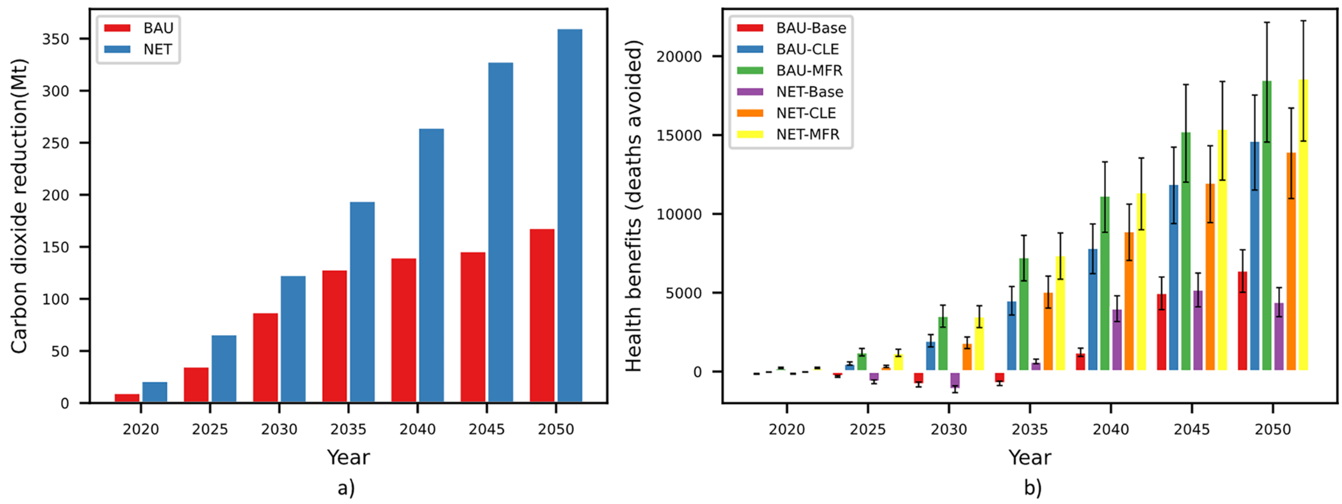

Figure 1. Total carbon dioxide reduction and health benefits under different electricity carbon mitigation targets (BAU and NET) and air pollution control (Base, CLE, and MFR) scenarios due to the ban policy in China. (a) Carbon dioxide reduction (Mt). (b) Health benefits (deaths avoided).

supported technology roadmap in the private vehicle sector.[41] The impacts of hydrogen fuel cell vehicles are not considered in this study. To easily observe the regional differences of the policy impacts, we set the same scenario in all provinces and the following discussion based on the discrepancy between BPS and REF (namely, BPS-REF). For the electricity sector, scenarios were designed according to the stringency of carbon mitigation targets and air pollution control in the MESEIC model. The business-as-usual (BAU) and net-zero (NET) scenarios are set for different carbon mitigation targets. Although the BAU and NET scenarios have the same projection for the future cost assumption of technologies, the BAU scenario does not consider the carbon cap, while the NET scenario is developed under the carbon neutrality goal that the carbon intensity will mainly follow the 2 °C target pathway but reduce more to reach near zero (a 97% reduction) by 2050.[2] In the air pollution control dimension, we considered baseline (BASE), current legislation (CLE), and maximum feasible reduction (MFR) scenarios to capture the uncertainties of the future level of end-of-pipe control for electricity generation. The BASE scenario assumes that the air pollutant reduction level remains the same as the base year. The CLE scenario represents the air pollutant reduction level approximately in line with the Beautiful China goal by 2035, corresponding to the national ambient air quality standards target of 35 μg/m^3. The MFR scenario requires the best available technologies to achieve the World Health Organization (WHO) Interim-3 standard of 15 μg/m^3 by 2050.[42]

2.2. Estimating Future Private Vehicle Fleet, Electricity Generation, and Transmission. To simulate the above scenarios, the CPPEE and MESEIC models, which run at a yearly interval from 2015 to 2050, were employed to estimate the future private vehicle (PV) fleet, electricity generation, and transmission. CPPEE is a bottom-up accounting model for estimating provincial transport demand, vehicle stock, energy demand, and CO_2 and air pollutant emissions in mainland China. The CO_2 and air pollutant emissions from private vehicles and increased electricity demand are the original outputs from CPPEE. MESEIC is a bottom-up provincial-level energy system model that can produce the lowest-cost technology mix for each province that fulfils both future electricity demand and CO_2 and air pollutant emission constraints.[36−38] Compared to other studies with a direct strong assumption in electricity scenarios, the future electricity technology mix and transmission can be modeled in MESEIC with its 101 transmission channels among all provinces in China in our study.

The quasi-input−output (QIO) model proposed by Qu et al.,[39] which is highly recognized by many studies,[33,43−45] was adopted to calculate the provincial CO_2 and air pollutant emissions for electricity consumption in the interconnected grid network. The input of QIO is the provincial electricity generation and demand for interprovincial electricity transmissions from the output of MESEIC. Combined with the electricity demand increased by private vehicles from outputs of CPPEE and the emissions for electricity consumed in a certain province from outputs of MESEIC and QIO, the indirect out-of-province emissions through electricity transmission from the ban policy would be revealed. More detailed descriptions of the models above are provided in Sections S1.1−S1.3 in the Supporting Information.

2.3. Estimating Provincial CO_2 and Air Pollutant Emissions. We calculated the CO_2, SO_2, NO_x, and $PM_{2.5}$ emissions caused by the ban policy of each province following the modeling results above. The total direct end-of-pipe CO_2 and SO_2 emissions from private vehicles are the product of the end-use energy consumption and emission intensity. For NO_x and $PM_{2.5}$, the total direct end-of-pipe emissions of private vehicles are the product of the total vehicle travel distance and the emission factor of air pollutants. The total indirect CO_2 and air pollutant emission changes in the electricity sector are the product of the total electricity consumption of EVs and the emission factor matrix, which represents the linkages of emission across provinces from the electricity consumed in a certain province. A more detailed description is provided in Section S1.4 in the Supporting Information.

2.4. Health Impact Assessment. We used an RCM (known as CHEER-AIR) to transform the emissions into atmospheric $PM_{2.5}$ concentrations at the province level.[46,47] The RCM is a more efficient method with a low cost of policy simulation and timely policy-making due to its simplicity and operability, which is the best choice in our study because thousands of scenarios need to be calculated. To calculate the avoided premature mortalities attributed to the reduced

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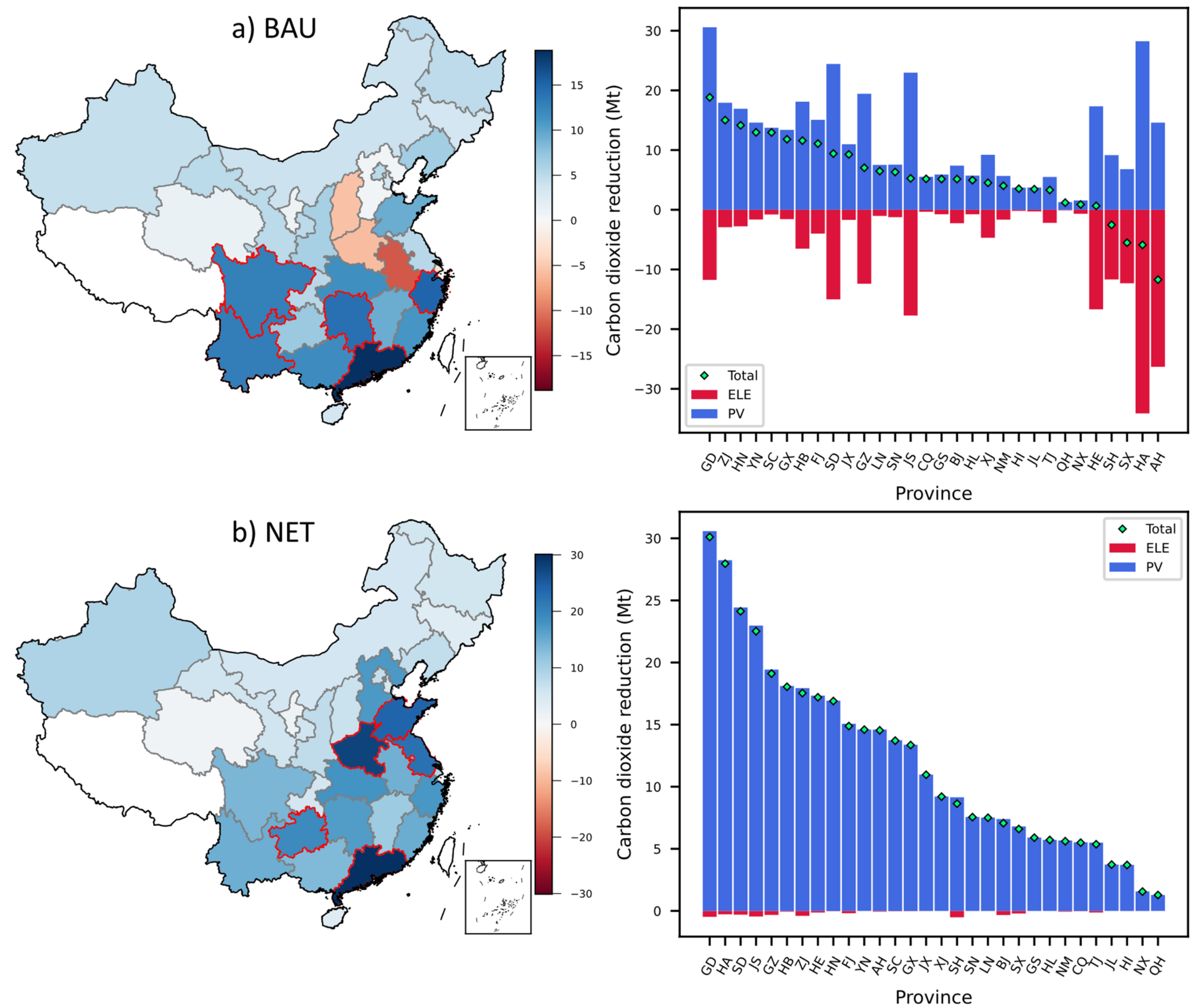

Figure 2. The carbon dioxide reduction (Mt) in 2050 due to a province-driven ban policy in different provinces under different carbon mitigation scenarios. (a) BAU. (b) NET. Note: The blue bar represents the carbon reduction from the PV sector, and the red bar represents the carbon emission from the electricity sector (ELE). The green dot represents the total carbon dioxide reduction, which has the same value as the map plot on left. The full names of the provinces hereinafter can be found in Table S27 in the Supporting Information.

ambient $PM_{2.5}$ concentration, we adopted The Global Exposure Mortality Model (GEMM).[40] A more detailed description is provided in Section S1.5 in the Supporting Information. The uncertainty of health benefits is considered in Section S2.3 in the Supporting Information.

2.5. Construction of the H/C Indicator. Referencing the method proposed by Li et al.,[10] the H/C indicator represents the health benefits (avoided premature mortality attributed to the reduced ambient $PM_{2.5}$ concentration) per amount of carbon reduction for the ban policy in each province. In this study, we estimated the health impacts (ΔM) and CO_2 emissions (ΔE_{CO_2}) changes in the preceding steps. Therefore, the H/C indicator due to the ban is directly calculated as $H/C = \Delta M/\Delta E_{CO_2}$.

3. RESULTS

3.1. Total CO_2 Reduction and Health Impact Changes Due to the Ban Policy under Different Electricity Scenarios.

In China, the differences in total CO_2 reduction and health impact due to the ban policy are tremendous under different electricity scenarios. The results of the fuel technology mix of private vehicle stock from CPPEE, the energy mix of electricity generation and electricity transmission from MESEIC, and the air pollutants emissions are provided in Figures S13−S18 and Section S2.1 in the Supporting Information. With the decarbonization of the electricity sector in China, the total CO_2 reduction under the NET scenario would be about twice as large as that under the BAU scenario (360.4 vs 168.5 Mt, in Figure 1a) in 2050. The differences in total CO_2 reduction between NET and BAU scenarios, increasing rapidly after 2035, follow the same time-varying pattern as the CO_2 reduction in the electricity sector (Figure S17). For health benefits (Figure 1b), the ban policy would avoid 6.4, 14.7, and 18.5 thousand cases of premature deaths in 2050 under the BAU-Base, BAU-CLE, and BAU-MFR scenarios, respectively. Under the NET scenario, the ban policy would avoid 4.4, 14.0, and 18.6 thousand cases of

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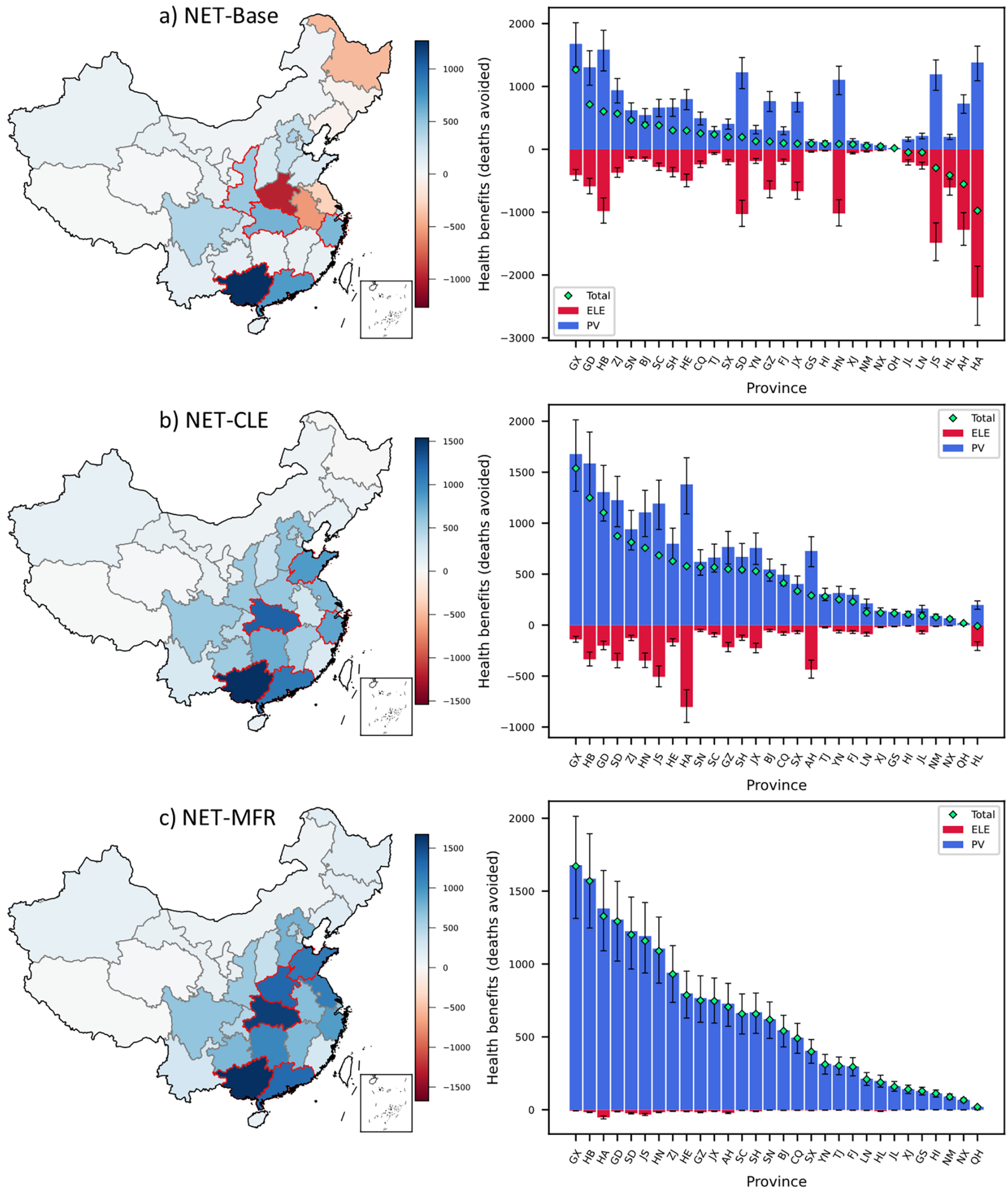

Figure 3. The health benefits (deaths avoided) in 2050 due to a province-driven ban policy in different provinces under NET with different air pollution control scenarios. (a) NET-Base. (b) NET-CLE. (c) NET-MFR. Note: The blue bar represents the health benefits from the PV sector, and the red bar represents the health damages from the electricity sector (ELE). The green dot represents the total health benefits, which has the same value as the map plot on left. The error bar represents the range of health benefits at the 95% CI.

premature deaths in 2050 under the NET-Base, NET-CLE, and NET-MFR scenarios, respectively. With the progress of air pollution control in the electricity sector, the gap in total health benefits between BAU and NET scenarios in China would narrow down. Based on the same time-varying pattern, the stronger the carbon mitigation targets are, the larger the potential health benefits are after 2035. Unlike this, the health benefits under the NET scenario are smaller than those under

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the BAU scenario in 2050 because of the increased use of biomass with its higher emission factor under the Base and CLE scenarios. With the strongest air pollution control (MFR), the total health benefits are always positive no matter which year and which carbon mitigation targets. The results indicate that the total CO_2 reduction and health benefits due to the ban policy are highly dependent on the electricity sector which required both carbon mitigation and air pollution control.

3.2. The Regional and Sectoral Distributions of CO_2 Reduction and Health Impact Changes Due to a Province-Driven Ban Policy. The regional and sectoral distributions of CO_2 reduction and health benefits due to a province-driven ban are greatly uneven. For regional differences, the top five provinces, mostly located in southeast and central China, would take up to over one-third of the total impact generally. Despite the CO_2 reduction and health benefits being always positive when only considering the PV sector, there is an opposite effect between PV and electricity sectors because the increased electricity consumption has the potential to cause a negative effect on CO_2 reduction and health benefits. As to CO_2 reduction (Figure 2), the top five provinces that can take up the highest amount of CO_2 reduction (i.e., Guangdong (GD), Zhejiang (ZJ), Hunan (HN), Yunnan (YN), and Sichuan (SC)) contribute to 44% of the total CO_2 reduction in China (or 18.8, 15.0, 14.1, 13.0, and 12.9 Mt CO_2 reduction, respectively) under the BAU scenario. Although there are CO_2 reductions due to the ban policy in most of the provinces considering both PV and electricity sectors, the CO_2 emissions due to the ban policy in Anhui (AH), Henan (HA), Shanxi (SX), and Shanghai (SH) provinces would increase because of the high carbon-intensive electricity they consume, while under the NET scenario, the top five provinces, four of which are not the same as those under BAU (i.e., Guangdong (GD), Henan (HA), Shandong (SD), Jiangsu (JS), and Guizhou (GZ)), contribute to 34% of the total CO_2 reduction in China (or 30.1, 28.0, 24.1, 22.5, and 19.1 Mt CO_2 reduction, respectively). The CO_2 emissions would decrease in all provinces in China under the NET scenario because the emission factor from the electricity sector would be very low and the increase in emissions in this sector would be negligible.

Regarding the health benefits, it is found that the regional and sectoral distributions of health benefits, dominated by different air pollution control actions, are similar under BAU (Figure S19) and NET (Figure 3) scenarios with the same level of emission intensity in electricity. Focused on the NET scenario, the ban policy in the top five provinces contributes 81%, 40%, and 38% of the total health benefits in China under the Base, CLE, and MFR scenarios. Air pollution control in the electricity sector would reduce regional inequities. Taking NET-CLE as an example (the most possible future), the top five provinces (i.e., Guangxi (GX), Hubei (HB), Guangdong (GD), Shandong (SD), and Zhejiang (ZJ)) would help avoid 1.54, 1.25, 1.10, 0.87, and 0.81 thousand cases of premature deaths for all of China, respectively. It is worth noting that not all the provinces would bring health benefits for all of China when banning ICEV sales under the NET-CLE scenario because of the large number of emissions from electricity consumed in a province such as Heilongjiang (HL), offsetting the health benefits gained from the transport sector. If the national average air pollutant emissions per unit of electricity generated could be reduced to 29.9% and 32.5% of that in 2015 (base year) under BAU and NET scenarios, respectively, an emission intensity level between the CLE and MFR scenarios, then the ban policy in every province would bring total health benefits for all of China by 2050 considering both PV and electricity sectors (Figure S20). With the strongest air pollution control in the electricity sector (under the NET-MFR scenario), the health damages in the electricity sector are negligible compared with the health benefits in the PV sector, and the distribution pattern of health benefits is similar to that in the NET-CLE scenario. Noteworthy, the ban policy in Henan (HA), Anhui (AH), Heilongjiang (HL), and Jiangsu (JS) would bring huge health damage to the whole of China under the NET-Base scenario because of the joint effect of the large proportion of coal-fired and biomass-fired electricity consumed in these provinces and the high emission factor. Under various carbon mitigation targets and air pollution control scenarios, the total health benefits for all of China when banning in Guangdong (GD), Guangxi (GX), and Hubei (HB) are relatively robustly high under different electricity scenarios due to their stable and clean energy mix and amount of electricity consumed. However, the total health benefits for all of China when banning in Anhui (AH), Henan (HA), Heilongjiang (HL), and Jiangsu (JS) vary most across scenarios. Moreover, the spillover effect brought by inter-regional atmospheric transportation and electricity transmission is non-negligible (the details are in Section S2.2). The indirect spillover emissions from electricity transmission are increased when the energy mix changes in electricity generation due to carbon neutrality. Despite the direct atmospheric transportation that would demonstrate the major regional spillover pattern, the indirect spillover effect from electricity transmission would increase the complexity of inter-regional-transferred health impact.

3.3. Provincial Ranking of the H/C Indicator. In total, health benefits per ton of carbon reduction (H/C, deaths avoided per million tons of carbon reduction) may vary by up to 8 times across provinces (Table S28). Accordingly, the ranking of H/C due to the ban policy was calculated, considering the impacts of only phasing out ICEVs in the private vehicle sector (PV only), as well as the combined impacts with the electricity sector under the NET scenario with different air pollution control actions (Base, CLE, and MFR) in 2050 (Table S29). Guangxi (GX), Chongqing (CQ), Shannxi (SN), Hubei (HB), Beijing (BJ), and Shanghai (SH) ranked stably high, implying that the ban policy in these provinces would bring more potential health cobenefits for each ton of carbon reduction, regardless of the decarbonization and air pollution control intensity of electricity generation. When designing the timetable for the ban on the sale of ICEVs in the private vehicle sector in China, these provinces are suggested to implement the ban policy first to gain more health benefits. However, the H/C values in Heilongjiang (HL), Henan (HA), Anhui (AH), Jiangsu (JS), Liaoning (LN), and Jilin (JL) change dramatically (over 100%), and these changes are more sensitive to electricity scenarios with low rankings under the Base scenario because of the high-polluting electricity consumed and high rankings under the MFR scenario. Therefore, the policy effect in these provinces relies heavily on the process of decarbonization and air pollution control in electricity generation.

To test the sensitivity of our results, we consider the uncertainties with the parameters of the CPPEE model and the transmission relationship in the MESEIC model. For health

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benefits calculation, the sensitivities from different E-R functions, baseline incidence rates of diseases, and background air-quality scenarios are considered in our study, as well. Almost all the values changed did not exceed the 95% confidence interval (CI) of the main analysis, demonstrating the robustness of our results. The internal standard error from parameters of exposure-response (E-R) functions, fully considered in advance, is the main contributor to the uncertainty in our results.[40,48] The detailed results of the sensitivity analysis are shown in Section S2.3 in the Supporting Information.

4. DISCUSSION

To incorporate health cobenefits into climate policy in sectors highly dependent on electrification for decarbonization, this study built an integrated assessment framework with energy, environment, and health models to quantify the carbon emission and health impact changes due to a province-driven ban on new sales of ICEVs in China. The results show that different electricity scenarios would change up to two-thirds of the CO_2 reduction and health benefits in the transport sector due to the ban policy. If there were no carbon neutrality and air pollution control goals in electricity generation, 53% of CO_2 reduction and 65% of health benefits would be offset by the increased power demand and consequent carbon emission and health damage in 2050. The regional and sectoral distributions of CO_2 reduction and health benefits are uneven. The top five provinces, mostly located in southeast and central China, would take up to over one-third of the total impact generally. It is found that air pollution control in the electricity sector would reduce the regional inequities in health benefits. If the national average air pollutant emissions per unit of electricity generated could be reduced a little bit more than that under the current policy (namely, 29.9−32.5% of the emission intensity in 2015), then the ban in every province would bring the total health benefits for all of China by 2050 considering both PV and electricity sectors. The results also reveal whether the health benefits gained from the ban in each province are sensitive to the electricity sector or not. Because the health benefits per ton of carbon reduction (H/C) may vary by up to 8 times across provinces, we ranked the province by H/C ratio to inform the suggested order to enact a province-driven ban policy to maximize health cobenefits in China.

It is worth noting that the H/C value changes dynamically year by year. Therefore, we also tested the H/C from 2020 to 2050, and the results show limited differences in provincial rankings compared to our main results in 2050 (Tables S29 & S30). The provinces are divided into three groups according to their volume and stability of H/C due to a province-driven ban from 2020 to 2050 under different electricity scenarios (the classification is in Table 2, the total health benefits are in Figure S21, and the H/C values under NET scenario are in Figure S22). The provinces in the "High" group have higher H/C values (mostly in southeast China and the Sichuan Basin), while provinces in the "Low" group have lower H/C values (mostly in west China). The interannual variation of H/C values in both High and Low groups are relatively stable across different electricity scenarios, indicating that the electricity sector contributes fewer impacts on the ban policy. However, the provinces in the "Change" group (mostly in central and northeast China) have dramatic H/C change across different electricity scenarios, mostly leading to health damage (negative H/C values) under Base scenarios in the

Table 2. Groups of Provincial Rankings for Developing Electric Vehicles or Banning the Sale of ICEVs in This Study and Other Studies[e]

province	our study	policy[a]	WRI[b]	Bu[c]	Wu[d]
GX	1	3	3	3	2
CQ	1	3	2	3	1
HB	1	2	2	2	1
SN	1	3	3	2	2
BJ	1	1	1	1	1
TJ	1	1	2	1	1
ZJ	1	1	1	1	2
SC	1	3	3	3	1
GD	1	1	1	1	1
HI	1	1	1	3	2
JX	2	2	2	3	2
SX	2	1	3	2	2
HE	2	1	2	2	2
NX	2	3	4	2	2
GS	2	3	4	3	2
YN	2	3	3	2	2
FJ	2	2	3	2	1
NM	2	3	4	2	2
QH	2	3	4	2	2
XJ	2	3	4	2	1
SH	3	1	1	1	1
HN	3	2	3	3	2
JS	3	1	2	1	1
SD	3	1	3	2	2
AH	3	2	2	3	2
HA	3	2	3	2	2
JL	3	3	4	2	2
GZ	3	3	2	3	2
HL	3	3	4	2	2
LN	3	3	4	1	2

[a]Notice on the "Thirteenth Five-Year" New Energy Vehicle Charging Infrastructure Incentive Policy and Strengthening the Promotion and Application of New Energy Vehicles.[49] [b]WRI, 2019.[22] [c]Bu et al., 2020.[23] [d]Wu et al., 2019[24] [e]The smaller the number is, the higher the group is ranked. For example, the High group is 1, the Low group is 2, and the Change group is 3. The font in bold indicates that the ranking of this study is quite different from that of others.

very first years or even in 2050, which would have a wide range of uncertainty when banning sales of ICEVs. To better compare the results under BAU and NET scenarios, the CO_2 reduction and health benefits curves are shown in Figures S23 & S24 because the CO_2 emissions in some provinces (most in the Change group) are increased under the BAU scenario and would be problematic if we still used H/C values (both negative CO_2 reduction and health benefits would result in positive H/C values). The curves also support our classification, as the amount and stability of CO_2 reduction and health benefits follow the same distributions. It is worth noting that the classifications of Low and Change are not set in stone because the ban policy in provinces in the Change group would have better performance with cleaner electricity. But no matter what the circumstances are, provinces in the High group are always encouraged to establish the ban policy first. Compared with the single values in 2050, despite Sichuan (SC), Guangdong (GD), and Hainan (HI) having lower H/C values, they should be encouraged to ban first because of the large amount of CO_2 reduction and stable health benefits. On the contrary, although Shanghai (SH) and Jiangsu (JS) have a higher H/C value, whether they would be encouraged to ban first is deeply dependent on the electricity sector because the health benefits change across years and scenarios. In our study, these rankings are more appropriate for the order of provincial policy implementation.

To further discuss the appropriate order of the ban policy, we compared the suggestions in our study with others that did not account for health cobenefits in the design of the ban policy (Table 2).[22−24] We found that current classifications are plausible to some extent. For Hubei (HB), Beijing (BJ), Tianjin (TJ), Zhejiang (ZJ), Guangdong (GD), and Hainan (HI), we suggest that these provinces should implement the ban policy first, the same as those results in previous studies. However, previous studies also suggest the prioritization of the implementation of this ban policy in Shanghai (SH) and Jiangsu (JS) with their developed economy, large population, and urgent demand to improve local air quality; they ranked low in our study due to the unstable health impacts under different electricity scenarios, which emphasized the efficiency of carbon mitigation in obtaining health cobenefits from a cross-sectoral perspective. Moreover, some provinces rank low in other studies but have high H/C values here, such as Guangxi (GX), Chongqing (CQ), Shannxi (SN), and Sichuan (SC), where implementing the ban policy would gain considerable CO_2 reduction and health benefits. However, the rankings in our study also imply that continuation of the current order of province-driven ban policy may miss the opportunity to maximize the health cobenefits of carbon reduction.

There are some limitations in our modeling framework that can be improved in the future. First, the boundary setting of our study can be expanded. Because of the lack of fine-resolution data, we focused only on PTW processes in private vehicles and electricity generation, neglecting the impacts of other fuel cycle and vehicle cycle processes. Other technological details such as the electricity loss by transmission and charging[50] and the benefits from vehicle-to-grid (V2G) technology[51,52] were not considered in our studies. Also, the provincial analysis could not capture the inner-province heterogeneity of health benefits. For example, population density is sparse around the power plant but dense in the urban area with heavy traffic, so the health benefits would be different even with the same emissions by assuming an evenly distributed population within a province.[53] Second, under limited computing resources, it is difficult to use the chemical transport model (CTM), a classic simulation method, to individually simulate perturbations to the concentration due to each provincial policy. Using the reduced complexity model here, although we simplified some chemical processes (e.g., the contribution from hydrocarbon), previous studies have used and validated this reliable tool.[10,46,47] Moreover, we could not include the ozone-related health impacts, which may also be important but no more than 5% of the $PM_{2.5}$-related health impacts in China.[15,19,54] Finally, the implementation order of the provincial ban on the sale of ICEVs is a complex issue considering not only climate and health cobenefits but also other regional impacts (physical health,[55] heavy metal pollution,[56] etc.) and realistic factors (economic development, cost,[57] infrastructure deployment,[58] zero-emission zone,[59] the will of stakeholders,[60] etc.). All of these multidimensional impacts and factors need to be compared under a uniform framework. These limitations could be improved by coupling with energy system models, impact assessment methods, or empirical economic analysis. Despite these limitations, our study provides valuable insights into how to incorporate health cobenefits into province-driven climate policy for sectors that highly depend on electrification for decarbonization.

In summary, our research provides a fresh perspective on the optimization of province-driven policy by capturing the policy impacts of each individual province instead of the nationally uniform policy. This modeling framework can also be used to quantify the impacts of similar policies in other sectors that highly depend on electrification for decarbonization, such as the industrial and residential sectors. By quantifying the contribution of the ban policy in different provinces, this study provides the cobenefits and trade-offs driven by each province and the scientific basis for which the province can first ban sales of ICEVs under consideration of health benefits. This approach is of practical significance for developing countries under the pressure of carbon reduction and severe air pollution challenges to incorporate health benefits into mitigation pathways.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.2c08450.

Description of model and method; relevant data source; results of the model output; supporting results of direct and indirect health impact transfer; sensitivity analysis; distribution of H/C values (PDF)

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Notes

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